

Thermodynamic Equilibrium

Henry's Law

It is assumed that the gaseous phase and the aqueous phase are in thermodynamic equilibrium. The equation for phase equilibrium is the equality of fugacities of the component in the gaseous and aqueous phase. i.e.

$$g_i = f_{ig} - f_{iw} = 0, i = 1, \dots, n_c \quad (\text{D2.1})$$

where

f_{ig} = fugacity of component i in the gas phase

f_{iw} = fugacity of component i in the aqueous phase

n_c = number of gaseous components

To be general, the gas is considered to be a multicomponent mixture. This is required if the injected CO₂ is impure or if flue gas (a mixture of CO₂ and N₂) is injected. The fugacity f_{ig} is calculated from an equation of state (EOS) ([Peng and Robinson, 1976](#), and [Soave 1972](#)). The fugacity f_{iw} of gaseous components soluble in the aqueous phase is calculated using Henry's law ([Li and Nghiem, 1986](#)), i.e.:

$$f_{iw} = y_{iw} \cdot H_i \quad (\text{D2.2})$$

where

H_i = Henry's constant of component i

y_{iw} = mole fraction of component i in the aqueous phase

The Henry's constants H_i are functions of pressure, temperature and salinity. In versions of GEM prior to 2008.10, Henry's constants at a given pressure, temperature and salinity are calculated by WinProp. The molar volume of the gaseous component in the aqueous solution is also estimated by WinProp at a given temperature. The following equation is then used to calculate Henry's constants at other pressures:

$$\ln H_i = \ln H_i^* + \frac{\bar{v}_i (p - p^*)}{RT} \quad (\text{D2.3})$$

where

H_i = Henry's constant for component i at pressure p and temperature T

H_i^* = Henry's constant for component i at pressure p^* and temperature T

p = pressure

p^* = reference pressure

R = universal gas constant

T = temperature (K)

$\bar{v}_i$ = partial molar volume of component i

The above approach assumes that the temperature and salinity do not change significantly throughout the aquifer and that a representative value for temperature and salinity could be used to calculate Henry's constants. This may not be applicable for a thick aquifer where the temperature difference between the top and bottom of the aquifer is substantial or for the case where CO₂ is injected at a temperature significantly different from that of the aquifer. In addition, the salinity of the aquifer may change due to water injection.

Thus an implementation of Henry's constant correlations will provide more flexibility to handle the above situations. [Harvey \(1996\)](#) published correlations for Henry's constants of many gaseous components including CO₂, N₂, H₂S and CH₄. The latter are particularly important in the sequestration of Greenhouse Gases (GHG). These correlations have been implemented in GEM and are described in the following.

Gas solubility depends on the salinity of the aqueous phase. The solubility of light gases normally decreases with increasing salinity, this phenomenon is referred to as the salting-out process. Models for salting-out coefficients have also been implemented in GEM.

Henry's Law Constant Correlations

The following correlation is due to [Harvey \(1996\)](#). Henry's constants at the H₂O saturation pressure and T are estimated from:

$$\ln H_i^s = \ln p_{\text{H}_2\text{O}}^s + A(T_{r,\text{H}_2\text{O}})^{-1} + B(1 - T_{r,\text{H}_2\text{O}})^{0.355} (T_{r,\text{H}_2\text{O}})^{-1} + C [\exp (1 - T_{r,\text{H}_2\text{O}})] (T_{r,\text{H}_2\text{O}})^{-0.41} \quad (\text{D2.4})$$

where

H_i^s = Henry's constant for component i at the saturation pressure of H_2O (MPa)

$p_{\text{H}_2\text{O}}^s$ = saturation pressure of H_2O in MPa at T (K)

$T_{c,\text{H}_2\text{O}}$ = critical temperature of H_2O (K)

$T_{r,\text{H}_2\text{O}}$ = $T/T_{c,\text{H}_2\text{O}}$, reduced temperature of H_2O

Table D-1 gives value of the parameters A , B and C for the gases of interest.

Table D-1: Parameters for Correlation of Aqueous Henry's Constants

Gaseous Solute	A	B	C
CO_2	-9.4234	4.0087	10.3199
N_2	-11.6184	4.9266	13.3445
H_2S	-5.7131	5.3727	5.4227
CH_4	-11.0094	4.8362	12.5220

The saturation pressure $p_{\text{H}_2\text{O}}^s$ of H_2O at T is calculated from the [Saul and Wagner \(1987\)](#) correlation equation. The Henry's law constant at p and T is then given by:

$$\ln H_i = \ln H_i^s + \frac{1}{RT} \int_{p_{H_2O}^s}^p \bar{v}_i dP \quad (D2.5)$$

where $\bar{v}_i$ is the partial molar volume of component i in the aqueous phase evaluated the T . The following correlations are used to estimate $\bar{v}_i$.

For CO₂, internally developed correlation that fits the experimental data from [Duan and Sun \(2003\)](#) is used:

$$\bar{v}_{CO_2} (cm^3/mol) = -47.75418 + 4.336154 \times 10^{-1} \times T - 5.945771 \times 10^{-4} \times T^2 \quad (D2.6)$$

where T is the temperature in K.

For CH₄, the correlation due to [Rettich et al. \(1981\)](#) is used:

$$\bar{v}_{CH_4} (cm^3/mol) = \exp \left(3.541 + 1.23 \times 10^{-3} \hat{T} \right) \quad (D2.7)$$

where $\hat{T}$ is the temperature in °C.

For N₂, the correlation due to [Perez and Heidemann \(2006\)](#) is used:

$$\bar{v}_{N_2} (cm^3/mol) = 15.732 + 6.60 \times 10^{-2} T \quad (D2.8)$$

For H₂S, internally developed correlation that fits Duan's data ([Duan et al. \(2007\)](#)) is used:

$$\bar{v}_{H_2S} (cm^3/mol) = 160.5567 - 5.538776 \times 10^{-1} T \quad (D2.9)$$

Effect of Salinity

The salting out coefficient implemented in GEM is defined by the following relation between the Henry's constant in pure water and the Henry's constant in brine ([Cramer, 1982](#)):

$$\log_{10} \left(\frac{H_{salt,i}}{H_i} \right) = k_{salt,i} m_{salt} \quad (D2.10)$$

where

$H_{salt,i}$ = Henry's constant of component i in brine (salt solution)

H_i = Henry's constant of component i at zero salinity

$k_{salt,i}$ = salting-out coefficient for component i

m_{salt} = molality of the dissolved salt (mol/kg H₂O)

[Bakker \(2003\)](#) gives the following correlations for the salting-out coefficients for CO₂ and CH₄ using data of [Cramer \(1982\)](#):

$$k_{salt,CO_2} = 0.11572 - 6.0293 \times 10^{-4} \hat{T} + 3.5817 \times 10^{-6} \hat{T}^2 - 3.7772 \times 10^{-9} \hat{T}^3 \quad (D2.11)$$

where $\hat{T}$ is the temperature in °C.

$$k_{salt,CH_4} = 3.38828 - 0.0318765 T + 1.22003 \times 10^{-4} T^2 - 2.31891 \times 10^{-7} T^3 \\ + 2.22938 \times 10^{-10} T^4 - 8.83764 \times 10^{-14} T^5 \quad (D2.12)$$

where T is the temperature in K.

[Perez and Heidemann \(2006\)](#) give the following correlation for the salting-out coefficient for N₂ and CH₄:

$$k_{salt,N_2} = 3.1150 - 0.01690 T + 2.4950 \times 10^{-5} T^2 \quad (D2.13)$$

Finally, [Suleimenov and Krupp \(1994\)](#) give the following correlation for the salting-out coefficient for H₂S:

$$\begin{aligned}
 k_{salt, H_2S} = & 8.37106265 \times 10^{-2} - 5.135608863 \times 10^{-4} T + 6.387039005 \times 10^{-6} T^2 \\
 & - 2.217360319 \times 10^{-8} T^3 - 5.069412169 \times 10^{-11} T^4 + 2.827486651 \times 10^{-13} T^5
 \end{aligned}
 \tag{D2.14}$$